

Supporting Information

Role of Carbonaceous Supports and Potassium Promoter on Higher Alcohols Synthesis over Copper-Iron Catalysts

Ho Ting Luk,[†] Cecilia Mondelli,^{,†} Sharon Mitchell,[†] Sebastian Siol,[‡] Joseph A. Stewart,[§] Daniel Curulla Ferré,[§] and Javier Pérez-Ramírez^{*,†}*

[†]Institute for Chemical and Bioengineering, Department of Chemistry and Applied Biosciences, ETH Zurich, Vladimir-Prelog-Weg 1, 8093 Zurich, Switzerland.

[‡]Empa, Swiss Federal Laboratories for Materials Science and Technology, Überlandstrasse 129, 8600 Dübendorf, Switzerland.

[§]Total Research & Technology Feluy, Zone Industrielle Feluy C, 7181 Seneffe, Belgium.

*Corresponding authors:

Cecilia Mondelli: cecilia.mondelli@chem.ethz.ch, Javier Pérez-Ramírez: jpr@chem.ethz.ch.

Table S1. Performance data for the catalysts studied.^a

Catalyst ^b	$WHSV$ ($\text{cm}^3 \text{g}_{\text{cat}}^{-1} \text{h}^{-1}$)	$STY_{\text{HA}[\text{ROH}]}$ ^b ($\text{g}_{\text{HA}} \text{g}_{\text{cat}}^{-1} \text{h}^{-1}$)	X_{CO} (%)	S_{MeOH} (%)	S_{HA} (%)	$S_{\text{HC=}}$ (%)	$S_{\text{HC-}}$ (%)	S_{CO_2} (%)
CNF-2	4000	0.13 [0.19]	48	7	20	7	49	17
	28000	0.23 [0.37]	8	13	31	18	32	6
CNF _p -2	4000	0.01 [0.03]	8	11	11	2	31	45
	12000	0.05 [0.08]	6	9	17	8	30	36
CNT-2	4000	0.07 [0.10]	35	5	15	2	62	16
	12000	0.06 [0.08]	7	3	21	14	53	9
AC	4000	n.r.	3	1	0	1	3	95
	500	n.r.	8	2	2	1	7	88
CP-2	4000	0.15 [0.16]	95	0.5	12	4	53	31
	48000	0.19 [0.21]	7	2	16	24	22	36
CNF-1	4000	0.06 [0.10]	49	4	9	8	54	25
	24000	0.15 [0.27]	8	8	21	17	31	23
CNF-3	4000	0.07 [0.13]	34	9	15	7	57	13
	24000	0.14 [0.32]	8	19	20	11	35	14
CNF-5	4000	0.08 [0.16]	46	9	13	4	64	10
	32000	0.12 [0.32]	8	18	16	9	41	16
CNF-2-C	4000	0.05 [0.09]	38	5	10	10	59	16
	16000	0.07 [0.13]	8	9	14	11	42	24
Fe/Cu/CNF-2	4000	0.08 [0.12]	29	5	12	12	50	21
	17000	0.05 [0.07]	8	7	16	18	27	32
CR-CNF-2	4000	0.02 [0.06]	22	9	8	3	46	34
	10000	0.02 [0.07]	8	13	9	5	38	35
DP-CNF-2	4000	0.04 [0.07]	23	8	12	5	63	12
	16000	0.05 [0.16]	8	17	12	4	32	35
CNF-2-0.1	4000	0.02 [0.02]	12	0.2	13	15	12	60
	8000	0.03 [0.04]	8	1	16	12	10	61
CNF-2-0.01	4000	0.09 [0.11]	44	3	15	12	17	53
	16000	0.22 [0.25]	8	6	47	15	14	18
CNF-2-0.005	4000	0.18 [0.26]	70	6	19	7	48	20
	32000	0.28 [0.42]	8	13	37	20	19	11
	32000 ^f	0.53 [0.67]	11	7	37	19	27	10
CuFe/K/CNF-2-0.005	4000	0.15 [0.21]	61	5	16	12	31	36
	20000	0.21 [0.28]	8	9	30	20	23	18
Cu/FeK/CNF-2-0.005	4000	0.08 [0.09]	37	2	14	12	17	55
	14000	0.11 [0.12]	8	3	26	17	25	29
Fe/CuK/CNF-2-0.005	4000	0.01 [0.02]	8	5	11	16	46	21
K/CuFe/CNF-2-0.005	4000	0.01 [0.02]	8	5	9	16	33	37
KCuFe/SiO ₂ -BM ^c	6000	0.23 [0.32]	53	23	38		11	28
CuFeMg ^d	2000	0.19 [0.28]	57	16	33		39	12
Cu ₂ Fe-3DOM ^e	2000	0.15 [0.19]	58	5	26		57	12

^aReaction conditions: 543 K, 5 MPa, and molar $\text{H}_2/\text{CO} = 2$. HA = higher alcohols, MeOH = methanol, HC = = olefins, and HC- = paraffins. All reported data are average values obtained between *ca.* 11–15 h on stream and the carbon balance is always higher than 95%. ^bCR = NaBH_4 reduction, DP = deposition precipitation, CP-2 or CP-5 = coprecipitation with Cu/Fe ratio of 2 or 5, SiO₂-BM = bimodal silica, n.r. = not relevant because negligible. ^cReference catalyst, data retrieved from *Fuel Process. Technol.* **2017**, 159, 436–441, reaction conducted at 593 K, 5 MPa, and molar $\text{H}_2/\text{CO} = 2$. STY_{HA} expressed in terms of $\text{g}_{\text{HA}} \text{cm}^3 \text{cat}^{-1} \text{h}^{-1}$. ^dReference catalyst, data retrieved from *Catal. Sci. Technol.* **2013**, 3, 1324–1332, reaction conducted at 573 K, 4 MPa, and molar $\text{H}_2/\text{CO} = 2$. STY_{HA} expressed in terms of $\text{g}_{\text{HA}} \text{cm}^3 \text{cat}^{-1} \text{h}^{-1}$. ^eReference catalyst, data retrieved from *ACS Catal.* **2017**, 7, 5500–5512, reaction conducted at 533 K, 4.8 MPa, and molar $\text{H}_2/\text{CO} = 1$. STY_{HA} determined based on the given HA formation rate and expressed in terms of $\text{mmol g}_{\text{cat}}^{-1} \text{h}^{-1}$. ^fReaction conditions for the long-term test: 548 K, 5 MPa, molar $\text{H}_2/\text{CO} = 1.5$, and $WHSV = 32000 \text{ cm}^3 \text{g}_{\text{cat}}^{-1} \text{h}^{-1}$. The reported data are average values obtained between 50–100 h on stream.

Table S2. Atomic concentrations of C, O, Fe, and Cu and surface Cu/Fe ratio for selected (K-promoted) CuFe catalysts supported on CNF in reduced and used forms.

Catalysts	C (at.%)	O (at.%)	Fe (at.%)	Cu (at.%)	Molar Cu/Fe ratio (-)
CNF-1	97.4	1.3	0.7	0.6	0.81
CNF-2	96.7	1.5	1.0	0.8	0.78
CNF-5	98.3	0.8	0.4	0.5	1.24
CR-CNF-2	81.8	9.4	6.8	2.0	0.29
CNF-2-0.005	96.4	1.9	0.8	0.9	1.22
CNF-2-0.005-used	98.5	1.0	0.4	0.1	0.31
K/CuFe/CNF-2-0.005	97.0	2.2	0.5	0.3	0.57

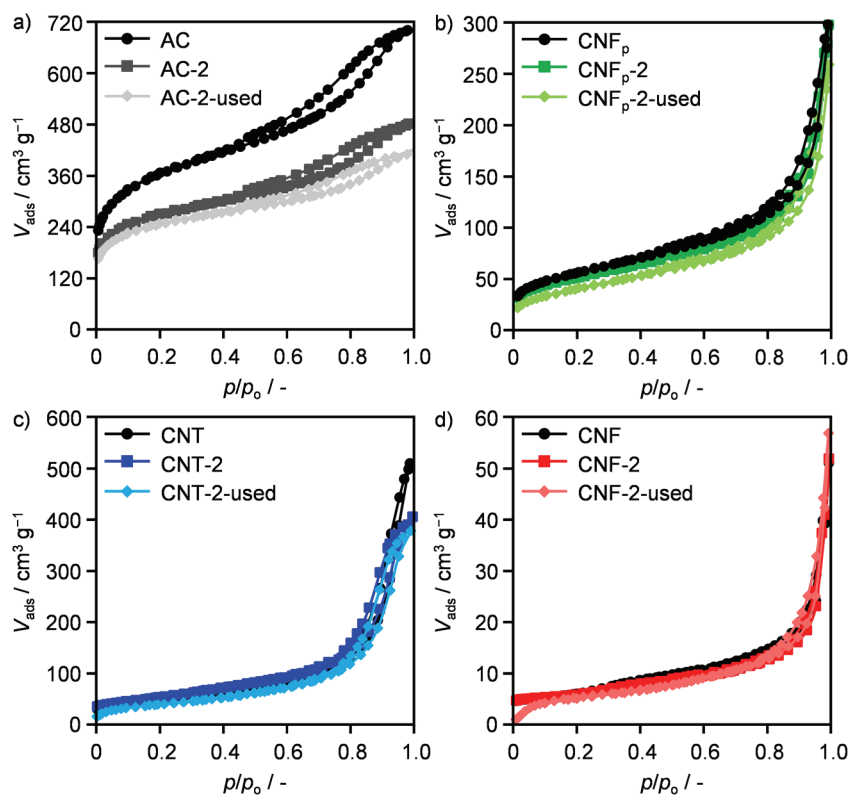


Figure S1. N_2 sorption isotherms at 77 K of a) AC, b) CNF_p , c) CNT, and d) CNF, and of the CuFe catalysts supported on these carriers in reduced and used forms.

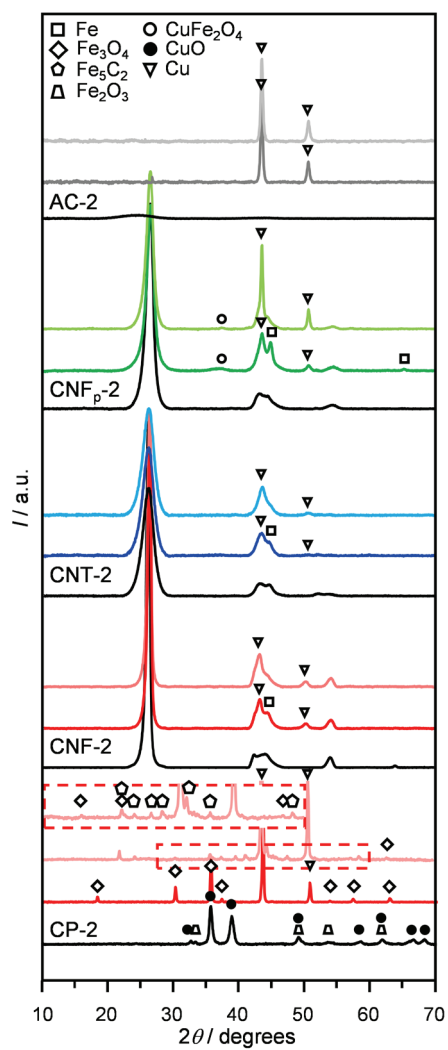


Figure S2. XRD patterns of bulk and carbon-supported CuFe catalysts in reduced (plain color) and used (pale color) forms. The patterns of the corresponding carriers and of CP-2 in calcined form are shown in black.

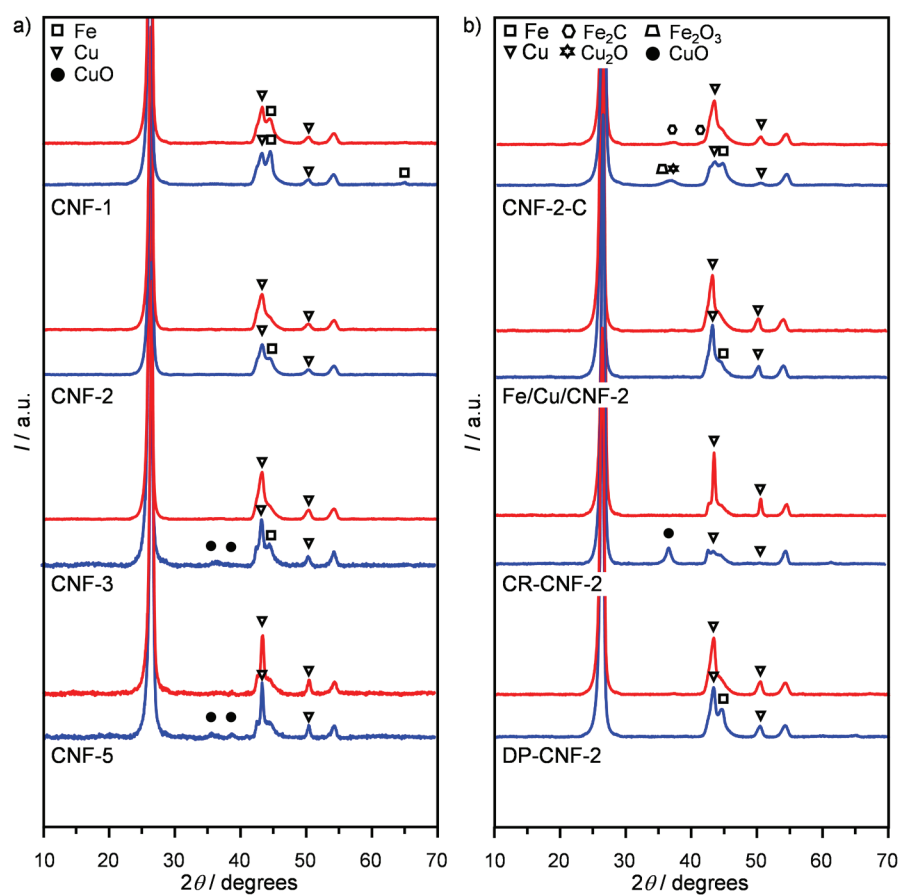


Figure S3. XRD patterns of CuFe catalysts supported on CNF and a) featuring different Cu/Fe ratios and b) prepared using different synthesis methods in reduced (blue), and used (red) forms.

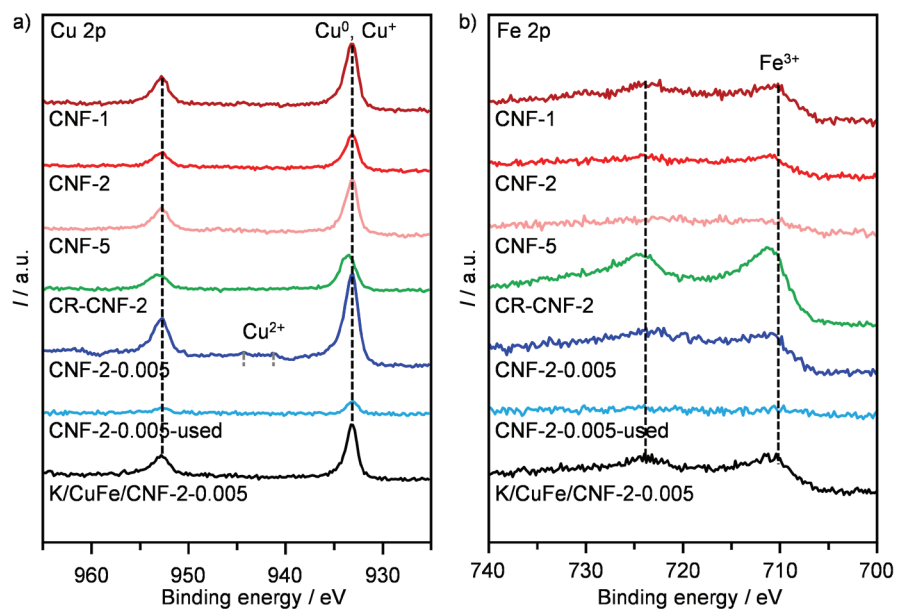


Figure S4. a) Cu2*p* and b) Fe2*p* core-level spectra of CuFe catalysts supported on CNF and featuring different Cu/Fe ratios or K as a promoter in reduced and used form.

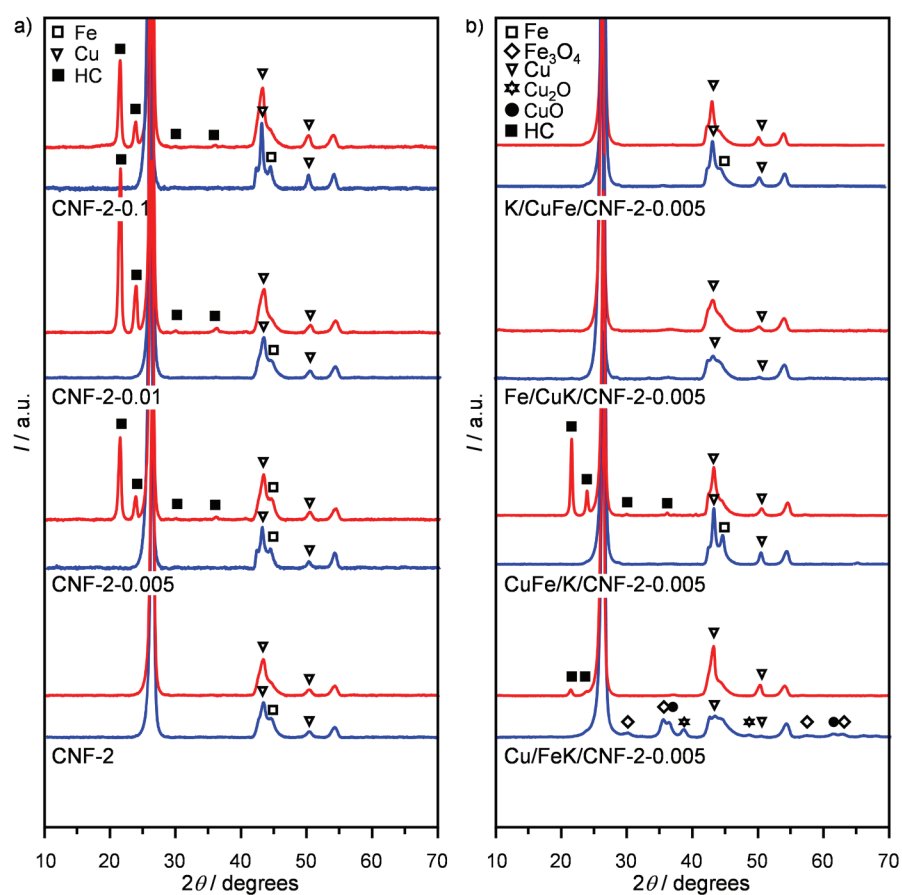


Figure S5. XRD patterns of K-promoted CuFe catalysts supported on CNF with a) distinct K contents, and b) added at different stages in the synthesis in reduced (blue) and used (red) forms.

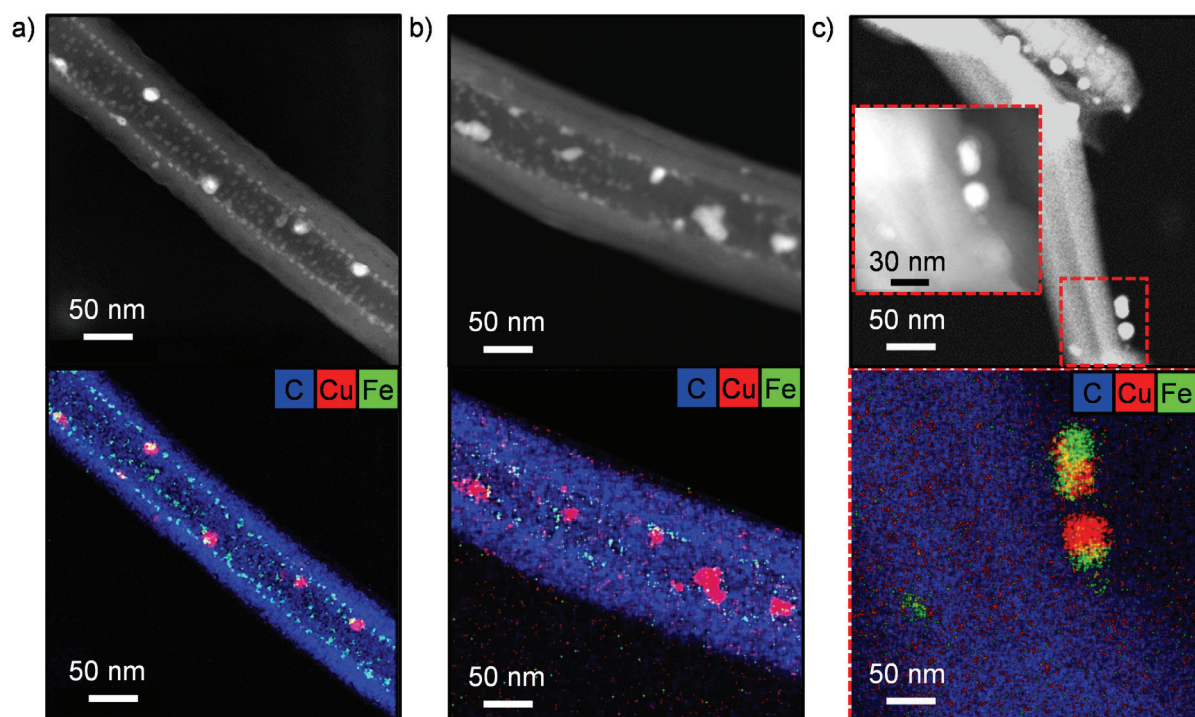


Figure S6. Additional STEM images and EDX maps of C (blue), Cu (red), and Fe (green) of a) CNF-2 and b) CNF-2-0.005 in reduced form as well as c) in used form.

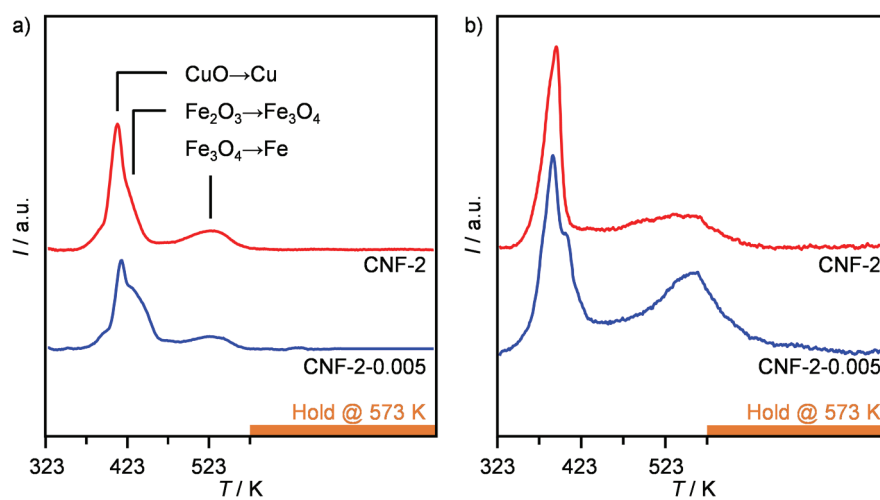


Figure S7. H₂-TPR profiles for CNF-2 and CNF-2-0.005 in reduced form measured at a) ambient pressure and b) 0.5 MPa prior to H₂-TPD analyses.

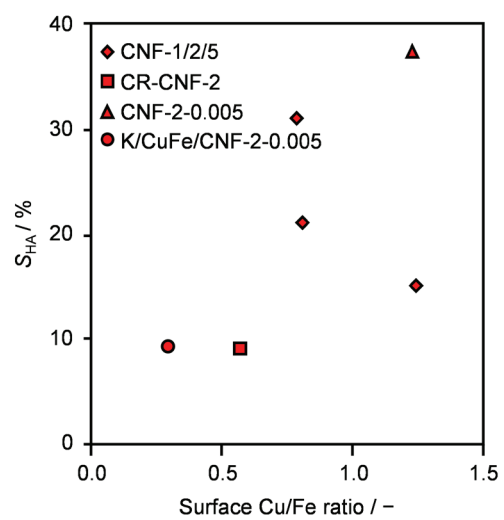


Figure S8. Selectivity to HA as a function of surface Cu/Fe ratio determined by XPS.

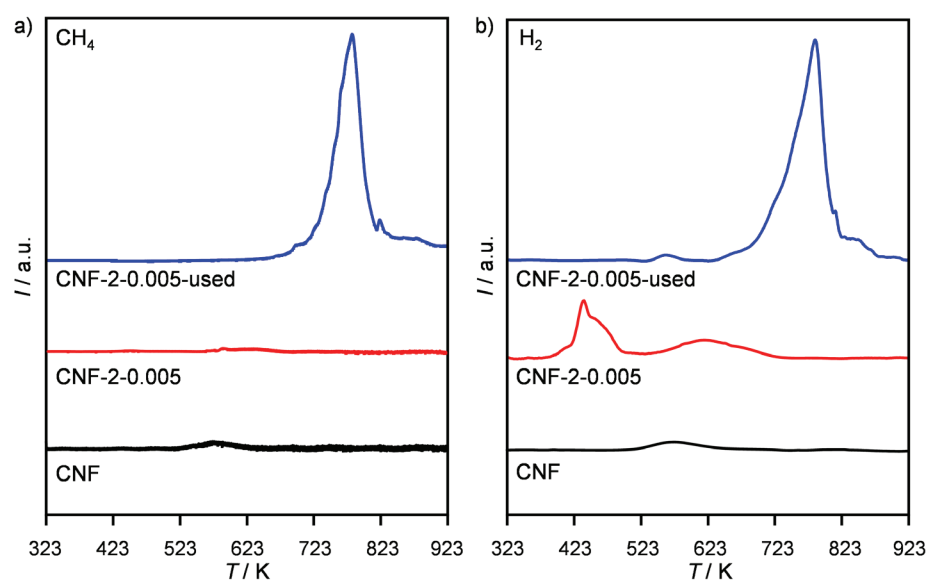


Figure S9. a) Methane formation and b) H₂ consumption by mass spectrometry ($m/z = 15$ and 2, respectively) upon the H₂-TPR analysis of CNF-2-0.005 in reduced and used forms and CNF.

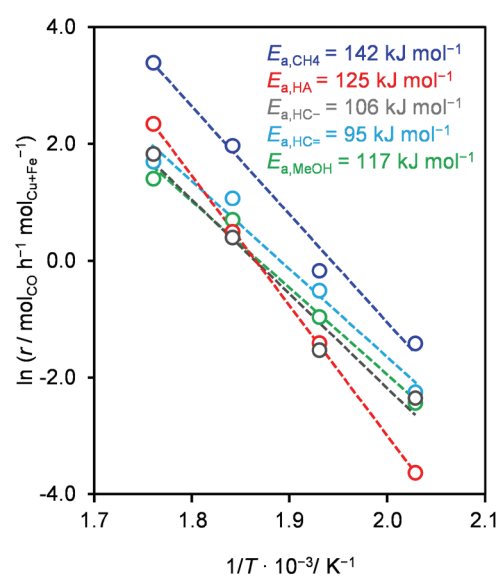


Figure S10. Arrhenius plot for CO hydrogenation to HA, methanol, methane, paraffins (excluding CH₄), and olefins over CNF-2-0.005.